Classical Machine Learning

Important Topics & Concepts for Interviews and Projects

1. ML Fundamentals

- What is Machine Learning?
- Supervised vs Unsupervised vs Semi-supervised Learning
- Classification vs Regression
- Features (X) vs Target (y)
- Training / Validation / Test sets
- Data leakage
- Underfitting vs Overfitting
- Bias vs Variance
- Generalization
- Baseline model

2. Data Preprocessing

- Missing-value handling
- Categorical encoding: One-Hot, Label, Ordinal Encoding
- Feature scaling: Standardization, Min-Max, Robust Scaling
- Outlier detection and handling
- Feature selection
- Feature engineering
- Train-test splitting
- Cross-validation
- Understand why preprocessing is needed, not only sklearn syntax.

3. Regression

- Linear Regression: equation, coefficients, loss function, MSE
- Gradient Descent and learning rate
- R^2 and Adjusted R^2
- Linear Regression assumptions
- Multicollinearity
- Regularization: Ridge (L2), Lasso (L1), Elastic Net

4. Classification

- Logistic Regression: sigmoid, probability, decision boundary, log loss
- Thresholds and multiclass classification
- KNN: distance metrics, choosing K, effect of scaling
- Naive Bayes: Bayes theorem, conditional probability, independence assumption
- SVM: hyperplane, margin, support vectors, kernel trick, C, gamma

5. Decision Trees

- Nodes, leaves and splitting
- Entropy
- Information Gain
- Gini Impurity
- Overfitting
- Pruning
- max_depth
- min_samples_split / min_samples_leaf

6. Ensemble Learning

- Bagging
- Random Forest: bagging, random feature selection, multiple trees, voting/averaging
- Boosting: weak learners focus on errors sequentially
- AdaBoost
- Gradient Boosting
- XGBoost
- LightGBM
- CatBoost

7. Model Evaluation

- Classification: Confusion Matrix, TP, TN, FP, FN
- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- PR-AUC
- Regression: MAE, MSE, RMSE, R^2, Adjusted R^2
- Know when to use each metric.

8. Cross-Validation

- K-Fold Cross-Validation
- Stratified K-Fold
- Leave-One-Out
- Why validation is needed
- Hyperparameter tuning without leaking the test set
- Workflow: training data → CV → choose model/hyperparameters → final test

9. Hyperparameter Tuning

- Parameters vs Hyperparameters

- Grid Search
- Random Search
- Bayesian Optimization: basic idea
- Examples: Random Forest (n_estimators, max_depth), KNN (n_neighbors), SVM (C, gamma), XGBoost (learning_rate, max_depth, n_estimators)

10. Unsupervised Learning

- K-Means: centroids, distance, assignment, update, convergence
- Choosing K
- Elbow Method
- Silhouette Score
- Hierarchical / Agglomerative Clustering
- Dendrogram and linkage
- DBSCAN: core point, border point, noise, eps, min_samples

11. Dimensionality Reduction

- PCA
- Why dimensionality reduction?
- Variance
- Principal components
- Eigenvectors/eigenvalues: basic intuition
- Explained variance
- Feature Selection vs Feature Extraction

12. Statistics for ML

- Mean, Median
- Variance and Standard Deviation
- Covariance and Correlation
- Probability
- Conditional Probability
- Bayes Theorem
- Normal Distribution
- Central Limit Theorem: basic understanding
- Sampling
- Confidence Intervals
- Hypothesis Testing
- p-value
- Statistical Significance

Most Important Algorithms

Priority: Linear Regression, Logistic Regression, Decision Tree, Random Forest, XGBoost, KNN, SVM, Naive Bayes, K-Means, PCA, Gradient Boosting.

Highest-Priority Interview Concepts

Focus first: Linear/Logistic Regression, Decision Trees, Random Forest, XGBoost, Bias-Variance, Overfitting/Underfitting, Feature Scaling, Cross-Validation, Evaluation Metrics, and Regularization.

The ML Pipeline to Understand

Raw Data → EDA → Train/Test Split → Preprocessing → Feature Engineering → Model → Cross-Validation → Hyperparameter Tuning → Evaluation → Error Analysis → Deployment